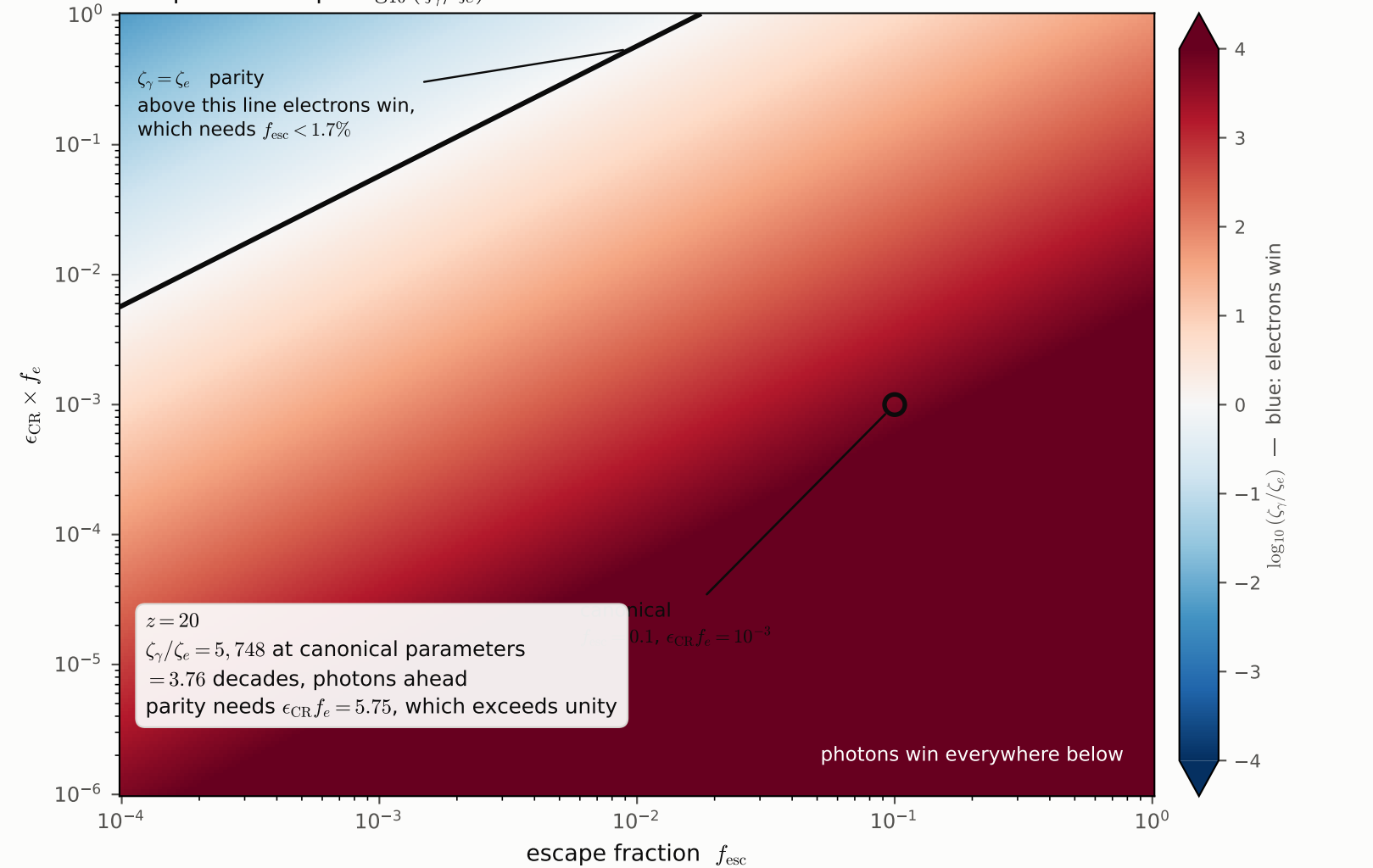


Ratio-only comparison at $z=20$ — independent of ρ_{SFR}
 competition map: $\log_{10}(\zeta_\gamma/\zeta_e)$



IGM at $z=20$, $x_e=10^{-4}$, pure H.
 Electron yields: the loss+IC route — igm_losses.py: 7 loss mechanisms, RBED secondary cascade, BED secondary cascade, Klein-Nishina IC, + IC-secondary photoionization.
 ρ_{UV} : Donnan+24 JWST PRIMER. $\xi_{\text{ion}}=10^{25.28}$: Llerena+25. $\rho_{\text{UV}} \rightarrow \rho_{\text{SFR}}$: Madau & Dickinson 14 (Salpeter).
 $\text{SN}/\text{M}_\odot$ derived from the IMF, not adopted.
 Stellar SED: population of U37126 ($z=10.255$, Marques-Chaves+26, BPASS v2.2.1, $Z=0.003$, 6.8 Myr); LyC shape parametrised, α bracketed 1–3.
 This figure contains NO absolute emissivity. ρ_{SFR} normalises the stellar photon budget and, through the supernova rate, the CR electron budget, so it cancels from the ratio exactly. The map is therefore quantitative at any redshift for which the yield physics holds, with no extrapolation of ρ_{UV} . Both axes are dimensionless fractions bounded by unity.